

FlowSentinel AI: GCP Setup, GitHub Issues/Projects & Data Strategy

Prepared by CTech Data Engineers for System & GitHub Administrators

🛠️ 1. GitHub Issues & Projects v2 Setup & Unblock Provisions

Required GitHub PAT Permissions (Fine-Grained Token):

Permission Scope	Access Level	Unblock Reason & Usage
issues	Read & Write	Auto-create incident tickets, assign labels, post RCA, and close issues on resolution.
pull_requests	Read & Write	Draft automated code fixes, link #Issue-ID , and request human review.
projects (v2)	Read & Write	Add incident cards to GitHub Projects board & set custom fields (Severity, Cost).
contents & workflows	Read & Write	Commit SQL patches and trigger GitHub Actions backfill workflows.

Secret Manager Command: `gcloud secrets create github-pat-token --data-file=`

📁 2. Required GCP APIs (13 Services)

Category	GCP API Service Identifier	Purpose in FlowSentinel AI
Orchestration & Compute	<code>composer.googleapis.com</code> , <code>run.googleapis.com</code>	Cloud Composer 3 (Airflow) & Cloud Run (React/FastAPI Ops Dashboard)
CI/CD & Containers	<code>cloudbuild.googleapis.com</code> , <code>artifactregistry.googleapis.com</code>	Docker builds for Cloud Run dashboard & custom Airflow operators
Data Warehouse & Lin	<code>bigquery.googleapis.com</code> , <code>dataform.googleapis.com</code>	BigQuery analytical engine, staging, & Dataform lineage graph parsing
Storage & Messaging	<code>storage.googleapis.com</code> , <code>pubsub.googleapis.com</code>	GCS landing buckets for vendor feeds; Pub/Sub incident event broker
AI, Observability & Sec	<code>aiplatform.googleapis.com</code> , <code>secretmanager.googleapis.com</code> , <code>logging.googleapis.com</code>	Vertex AI (Gemini 2.0) agentic triage, Secret Manager, & Cloud Logging sinks

🔑 3. GCP Service Account & IAM Roles Matrix

Service Account: `flowsentinel-agent-sa@<project-id>.iam.gserviceaccount.com`

IAM Role Name	Role Identifier	Reason / Access Scope
BigQuery Admin	<code>roles/bigquery.admin</code>	Querying execution logs, schema definitions, running backfill jobs
Storage Object Admin	<code>roles/storage.objectAdmin</code>	Managing GCS landing buckets for 3rd-party vendor CSV/JSON feeds
Cloud Composer User	<code>roles/composer.user</code>	Triggering Airflow task retries & inspecting XCom states
Vertex AI User	<code>roles/aiplatform.user</code>	Executing Gemini 2.0 agentic triage & code repair prompts
Secret Accessor	<code>roles/secretmanager.secretAccessor</code>	Retrieving <code>github-pat-token</code> and Slack Webhooks

📦 4. Data Provisioning & Anomaly Injector Strategy

Architecture: GCS Landing Buckets (`vendor_ecom/` , `vendor_crm/`) → BigQuery Datasets (`raw_staging` , `dw_analytics` , `de_ops_metadata` , `backfill_sandbox`) → **Python Data Injector CLI** (Simulates live Schema Drift, Feed SLA Delays, and Metric Formula Diffs).

Admin Action Checklist:

- [] Create GCP Project `ctech-flowsentinel-demo-dev` & Enable 13 APIs
- [] Create GitHub Repo `ctech-flowsentinel-demo` & Enable Projects v2
- [] Issue GitHub PAT with `issues` , `pull_requests` , `projects` , `contents` , `workflows` scopes
- [] Add GitHub PAT token to GCP Secret Manager as `github-pat-token`
- [] Create Service Account `flowsentinel-agent-sa` & Assign IAM roles